NFTmarket

Solving two fundamental gaps in the NFT ecosystem

AI Art • Watermarking • NFTs • Legal Licensing

The Two Problems with NFTs

Problem 1: No Legal Rights

NFT purchase doesn't transfer copyright

"Owning the NFT" ≠ owning the art

Problem 2: Hosting Fragility

NFTs depend on external hosting (IPFS)

If hosting disappears, NFT points to nothing

Solution 1: Explicit Licensing

Personal

Non-commercial display only

Commercial

Monetization rights

Full Transfer

Complete copyright transfer

Buyer's wallet signs license contract at point of sale.

License agreements stored permanently on Arweave.

Solution 2: Invisible Watermarking

- Information embedded in image frequency domain
- Imperceptible to humans, detectable by software
- Same technique Google uses for AI image detection (SynthID)

Embedded data:

- Creator's wallet address
- Token ID

Survives compression, resizing, format conversion.

Why Hosting Fails

- IPFS files only exist if someone is hosting them
- Pinning services require ongoing payment
- Platform stops paying → images disappear

Tokens remain on blockchain forever — pointing to nothing.

Proving Ownership Without IPFS

1. Extract Watermark

Reveals: creator wallet + token ID

2. Query Blockchain

Look up token ID → get current owner

The image carries its own proof of origin.

If Hosting Fails

Any surviving copy can prove:

Origin

Extract watermark → creator wallet + token ID

Current Ownership

Query blockchain with token ID

Built-in Marketplace

Standard ERC-721:

- External marketplaces, no licensing

NFTmarket:

- list(), buy(), delist() in one contract
- License terms attached to every transaction
- Rules enforced by code, not platforms

Limitations

- Other marketplaces can implement NFTmarket-compliant trading
- But selling on non-compliant platforms bypasses licensing

Single Point of Failure

- Watermark detection service is centralized
- If NFTmarket goes offline, extraction unavailable

Summary

Problem: No legal rights

Solution: Explicit licensing on Arweave

Problem: Hosting can disappear

Solution: Watermark embeds token ID + wallet

If one copy survives: origin and ownership can be proven.